ALL DRAINAGE TO BE CONCEALED IN DUCTS.

P: DRAINAGE:

1. ALL JOINTS BETWEEN PIPES SHALL BE APPROPRIATE TO AND COMPATIBLE WITH THE MATERIALS OF WHICH SUCH PIPES AND FITTINGS ARE MADE.
2. ALL JOINTS SHALL REMAIN WATERTIGHT UNDER NORMAL WORKING CONDITIONS.
3. JOINTS SHALL BE ABLE TO WITHSTAND AN INTERNAL WATER PRESSURE OF 50KPA AND AN EXTERNAL WATER PRESSURE OF 30KPA.
4. ANY SANITARY FIXTURE SHALL DISCHARGE THROUGH A TRAP INTO A SOIL PIPE OR WASTE PIPE, AS THE CASE MIGHT BE.
5. THE WATER SUPPLY OUTLET TO ANY WASTE FIXTURE SHALL BE SITUATED NOT LESS THAN 20MM ABOVE THE FLOOD-LEVEL RIF OF SUCH FIXTURE.
6. CLOTHES WASHING MACHINES OR DISH WASHING MACHINES WHICH IS PERMANENTLY CONNECTED TO A DRAINAGE INSTALLATION SHALL DISCHARGE THROUGH A TAP INTO A WASTE PIPE.
7. TO BE COMPLETED...

V: SPACE HEATING:

1. THE CHIMNEY OR FLUE OF AN OPEN SOLID-FUEL BURNING APPLIANCE (BURNING TIMBER OR COAL) SHALL BE PROVIDED WITH A DAMPER OR FLAP THAT CAN BE CLOSED TO SEAL THE CHIMNEY OR FLUE.
2. NO FLUE PIPE SHALL BE DESIGNED AND INSTALLED IN SUCH A MANNER THAT IT WILL CAUSE A FIRE HAZARD.
3. NO FLUE PIPE SHALL BE CONNECTED TO A SHAFT OR DUCT WHICH FORMS PART OF A VENTILATION SYSTEM.
4. COMBUSTIBLE MATERIAL SUCH AS A TIMBER FLOOR JOIST, TIMBER OR ROOF TRUSS, SHALL NOT BE BUILT WITHIN 200MM OF THE INSIDE OF A CHIMNEY.
5. CHIMNEYS SHALL BE MADE OF 220MMTHICK MASONRY UNITS.
6. WHERE A CHIMNEY IS PROVIDED WITH A FLUE LINING, SUCH LINING SHALL BE MADE OF A MATERIAL WHICH WILL WITHSTAND ANY ACTION OF THE FLUE GASSES AND WILL RESIST, WITHOUT CRACKING OR SOFTENING, THE TEMPERATURES TO WHICH IT MIGHT BE SUBJECTED, AND IT SHALL EXTEND THOUGHT THE FULL HEIGHT OF SUCH A CHIMNEY.

F: SITE OPERATIONS:

1. BEFORE ANY FOUNDATION IS LAID, THE AREA TO BE COVERED BY ANY BUILDING SHALL BE PROPERLY CLEARED OF VEGETABLE MATTER, TREE STUMPS, TIMBER AND OTHER CELLULOSE MATERIAL, DEBRIS OR REFUSE AND ANY MATERIAL CONTAMINATED WITH FECAL MATTER.
2. WHERE ANY SITE UPON WHICH A BUILDING IS TO BE ERECTED IS WATERLOGGED, SEASONALLY WATERLOGGED OR SATURATED, OR WHERE ANY BUILDING IS TO BE SO SITUATED THAT WATER WILL DRAIN NATURALLY TOWARDS IT, DRAINAGE SHALL BE PROVIDED TO DIRECT SUCH WATER AWAY FROM THE SITE OR BUILDING TO A STORM-WATER DRAIN OR DISPOSE OF I IN SOME OTHER SAFE APPROVED MANNER.
3. ALL STORMWATER DISPOSAL ARRANGEMENTS DURING CONSTRUCTION SHALL COMPLY WITH THE REQUIREMENTS OD SANS 10400-R.
4. THAT PART OF ANY DRAINAGE INSTALLATION OUTSIDE A BUILDING AND WHICH IS BELOW GROUND LEVEL, BUT SHALL NOT INCLUDE THE FOLLOWING: ANY DISCHARGE PIPE: THAT PORTION OF A DISCHARGE STACK WHICH IS BELOW GROUND LEVEL: THE BEND AT THE FOOT OF A DISCHARGE STACK
5. THAT PART OF A BUILDING WHICH IS IN DIRECT CONTACT WITH AND IS INTENDED TO TRANSMIT LOADS TO THE GROUND.
6. WATER RESULTING FROM NATURAL PRECIPITATION OR ACCUMULATION AND INCLUDES RAINWATER, SURFACE WATER, SUBSOIL WATER OR SPRING WATER.
7. PIPE, CONDUIT OR SURFACE CHANNEL SITUATED ON A SITE, WHICH IS USED TO CONVEY STORMWATER TO A SUITABLE POINT OF DISCHARGE.

J: FLOORS:

1. INSULATION SHALL BE INSTALLED AROUND THE EDGE OF THE WALL'S PERIMETER.
2. THE INSULATION SHALL HAVE AN R-VALUE OF MIN. 1,0.
3. FLOORS IN ANY LAUNDRY, KITCHEN, SHOWER ROOM, BATHROOM OR ROOM CONTAINING A TOILET PAN OR URINAL SHALL BE WATERPROOFED ON TOP.
4. FLOORS SHALL HAVE A FIRE RESISTANCE APPROPRIATE TO ITS USE AND WHERE REQUIRED, BE NON-COMBUSTIBLE.

K: EXTERNAL WALLS:

1. THE DESIGN OF THE EXTERNAL WALLS TO COMPLY WITH THE MINIMUM REQUIREMENT OF SANS 10400-ZA:2011 FOR EXTERNAL WALLS

M: STAIRWAYS:

1. RISERS TO BE 1660MM HIGH AND TREADS TO BE 300MM WIDE AND GOING TO BE 350MM WIDE.
2. THE RISE OF ANY STEP SHALL NOT EXCEED 200MM.
3. THE WIDTH OF ANY STAIR SHALL NOT BE LESS THAN 250MM PROVIDED THAT WHERE THE STAIRWAY DOES NOT HAVE SOLID RISERS, EACH TREAD SHALL OVERLAP THE NEXT LOWER TREAD BY NOT LESS THAN 25mm.
4. THE HANDRAIL TO ANY FLIGHT OF STAIRS PROVIDED ON AT LEAST ONE SIDE OF THE FLIGHT WHERE THE WIDTH OF THE FLIGHT IS LESS THAN 1.0m AND ON BOTH SIDES WHERE THE WIDTH EXCEEDS 1.1m SECURELY FIXED TO SUCH WALL, SCREEN, RAILING OR BALUSTRADE AT A HEIGHT OF NOT LESS THAN 850mm AND NOT MORE THAN 1m MEASURED VERTICALLY FORM THE PITCH LINE TO THE UPPER SURFACE OF THE HANDRAIL AND OF A DESIGN AND SO FIXED THAT THERE SHALL BE NO OBSTRUCTIONS ON, ABOVE, OR NEAR TO IT WHICH MIGHT OBSTRUCT MOVEMENT OF A HAND MOVING ALONG IT.

N:GLAZING AND FENESTRATION:

1. AIR LEAKAGE, CONDUCTION AND SOLAR HEAT GAIN IS CALCULATED DURING THE DESIGN OF THE HOUSE CONSIDERING NATURAL VENTILATION AND DESIGNED OPTIMALLY WITHIN THE SANS10400 REQUIREMENTS.
2. THE ONUS IS ON THE OWNER TO INFORM THE ARCHITECT IF ANY REVISIONS TO THE LAYOUT IS PLANNED IN ORDER FOR RECALCULATION TO BE DONE.
3. DIMENSIONS FOR VERTICAL GLASS SUPPORTED BY A FRAME ON ALL SIDES IN EXTERNAL WALLS IN BUILDINGS WHERE THE HEIGHT MEASURED FROM THE GROUND TO THE TOP OF SUCH WALLS DO NOT EXCEED 10m:
4. FOR ANY OTHER APPLICATION NOT MENTIONED ABOVE, CONSULT WITH THE ARCHITECT.
5. WHERE TRANSPARENT GLAZING IS USED AND IS NOT LIKELY TO BE APPARENT TO, OR SUSPECTED BY, ANY PERSON APPROACHING IT, SUCH GLAZING SHALL BEAR MARKINGS THAT SHALL RENDER IT APPARENT TO SUCH PERSON.

HOT WATER SUPPLY:

1. GAS HEATING SYSTEM TO COMPLY WITH SANS 1307, SANS 10106, SANS 10254 AND SANS 10252-1.
2. REQUIREMENTS FOR WATER INSTALLATIONS IN BUILDINGS SHALL BE IN ACCORDANCE WITH SANS 10252-1 AND SANS 10254
3. ALL HOT WATER SERVICE PIPES SHALL BE CLAD WITH INSULATION WITH A MINIMUM R-VALUE IN ACCORDANCE WITH SANS 204XA TABLE 1.
4. THERMAL INSULATION, SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS.
5. PIPING TO BE INSOLATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING SYSTEM. WHERE POSSIBLE, LENGTHS OP PIPE RUNS SHOULD BE MINIMIZED.

GLAZING:

1. THE INSTALLATION OF GLAZING IN BUILDINGS TO COMPLY WITH SANS 10137
2. GLAZING TO BE SUPPLIED & FITTED BY A COMPETENT PERSON WHO IS RECOGNIZED BY AN INSTITUTE, WHICH HAS SPECIALIST EXPERTISE IN THE FIELD OF GLAZING, AS GENERALLY HAVING THE NECESSARY EXPERIENCE AND TRAINING TO DETERMINE GLAZING REQUIREMENTS IN ACCORDANCE WITH THE REQUIREMENTS OF SANS 10137
3. SAFETY GLAZING MATERIAL TO COMPLY WITH THE REQUIREMENTS OF SANS 1263-1 FOR THE PERFORMANCE OF SAFETY GLAZING MATERIALS AND IN ACCORDANCE WITH SANS. 10137 CODE OF PRACTICE - FRAMES TO RECEIVE GLAZING MATERIAL SHALL EITHER COMPLY WITH THE REQUIREMENTS OF SANS 727 OR SANS 1553-2, OR BE CAPABLE OF WITHSTANDING THE WIND AND IMPACT LOADS DETERMINED IN ACCORDANCE WITH THE REQUIREMENTS OF SANS 10400-B WITHOUT DEFLECTING MORE THAN 1/175TH OF THEIR SPAN. ALL SAFETY GLAZING TO SANS. 1263 - PART 1 (PREVIOUSLY SABS.) AND IN ACCORDANCE WITH SANS. 10137 CODE OF PRACTICE AND ISO 9050 (GLASS IN BUILDING - DETERMINATION OF LIGHT TRANSMITTANCE, SOLAR DIRECT TRANSMITTANCE, TOTAL SOLAR ENERGY TRANSMITTANCE, ULTRAVIOLET TRANSMITTANCE AND RELATED GLAZING FACTORS) GLAZING - PERMISSIBLE AIR LEAKAGE (AL)
4. MAXIMUM PERMISSIBLE AL FOR OPENABLE GLAZING SHALL BE 2,0 L/S · M2 WITH A PRESSURE. DIFFERENCE OF 75 PA, WHEN TESTED IN ACCORDANCE WITH SANS 613.
5. MAXIMUM PERMISSIBLE AL FOR NON-OPENABLE GLAZING SHALL BE 0,31 L/S · M2 WITH A PRESSURE DIFFERENCE OF 75 PA, WHEN TESTED IN ACCORDANCE WITH SANS 613. FOR GLAZED DOUBLE ACTION SWING DOORS AND REVOLVING DOORS, THE MAXIMUM PERMISSIBLE AL SHALL BE 5,0 L/S · M2 WITH A PRESSURE DIFFERENCE OF 75 PA, WHEN TESTED IN ACCORDANCE WITH SANS 613. EXTERNAL DOORS - PERMISSIBLE AIR LEAKAGE (AL)
6. A SEAL TO RESTRICT AL SHALL BE FITTED TO EACH EDGE OF AN EXTERNAL DOOR AND OTHER SUCH OPENING THAT
   A) SERVES A CONDITIONED SPACE, OR
   B) SERVES A HABITABLE ROOM THE SEAL MAY BE A FOAM OR RUBBER COMPRESSIBLE STRIP OR A FIBROUS SEAL.
7. EXTERNAL SWING DOORS SHALL BE FITTED WITH A DRAUGHT PROTECTION DEVICE TO THE BOTTOM EDGE ON EACH LEAF

X: ENVIRONMENTAL SUSTAINABILITY: :

ROOF LIGHTS AND SKYLIGHTS

1. SHALL BE SEALED, OR CAPABLE OF BEING SEALED.
2. SHALL BE CONSTRUCTED WITH A COMPRESSIBLE SEAL IF THEY ARE ABLE TO OPEN.

EXTERNAL DOORS:

1. SHALL BE FITTED WITH A SEAL TO EACH EDGE OF AN EXTERNAL DOOR.
2. THE SEAL SHALL BE FOAM RUBBER COMPRESSIBLE STRIP OR FIBROUS SEAL I.E. VELT
3. EXTERNAL SWING DOORS SHALL BE FITTED WITH A DRAUGHT PROTECTION DEVICE AT THE BOTTOM EDGE ON EACH LEAF.

HOT WATER SERVICES:

1. A MINIMUM OF 50% BY VOLUME OF THE ANNUAL AVERAGE HOT WATER HEATING REQUIREMENT SHALL BE PROVIDED BY MEANS OF OTHER THAN ELECTRICAL RESISTANCE HEATING INCLUDING BUT NOT LIMITED TO SOLAR HEATING, HEAT PUMPS, HEAT RECOVERY FROM OTHER SYSTEMS OR PROCESSES.
2. ALL EXPOSED PIPES TO AND FROM THE HOT WATER CYLINDERS AND CENTRAL HEATING SYSTEMS SHALL BE INSULATE WITH PIPE INSULATED MATERIAL WITH THE FOLLOWING R-VALUES: <80mm INTERNAL [=MIN 1.0 R-VALUE AND >80mm [INTERNAL=MIN 1.5 R-VALUE.
3. INSULATION SHALL BE PROTECTED AGAINST EFFECTS OF THE WEATHER AND SUNLIGHT, BE ABLE TO WITHSTAND THE TEMPERATURES WITHIN THE PIPING AND ACHIEVE THE ABOVE MENTIONED R-VALUES.
4. HOT WATER VESSELS AND TANKS SHALL BE INSULATED WITH A MATERIAL ACHIEVING A MIN. R-VALUE OF 2.0.
5. INSULATIONS ON VESSELS, TANKS AND PIPING CONTAINING COOLING WATER SHALL BE PROTECTED BY A VAPOR BARRIER ON THE OUTSIDE OF THE INSTALLATION.
6. PIPING TO BE INSULATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM AND PRESSURE RELIEF PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR SYSTEM.

STOVES AND FIREPLACES:

- PROVIDE AN AIR SUPPLY FROM THE OUTSIDE THAT DOES NOT COOL THE INSIDE AIR.
- PROVIDE A DAMPER TO THE CHIMNEY THAT BE WHEN THE FIREPLACE IS NOT IN USE

NATURAL LIGHTING

10% OF THE FLOOR AREA OF THE ROOM(S), OR 0,2 m², WHICHEVER IS THE GREATER.

NATURAL VENTILATION

THE TOTAL AREA OF AN OPENING, A DOOR OR AN OPENABLE GLAZED WINDOW SHALL BE NOT LESS THAN 5 % OF THE FLOOR AREA OF THE ROOM, OR 0,2m² WHICHEVER IS THE GREATER.

BUILDING SEALING:

1. SEAL ALL GAPS WHERE CONDUITS, ELECTRICAL BOXES, PIPES AND DUCTS PENETRATE, USING AN ELASTIC SEALER LIKE FOAMED POLYURETHANE. THE CONTACT SURFACES SHOULD BE MOISTENED BEFORE APPLICATION OF THE SEALANT.
2. SEAL THE GAP BETWEEN THE ROOF AND THE OUTSIDE WALL, TAKING CARE THAT THE ROOF MOVEMENT DOES NOT BREAK AWAY THE FILLING.

TIMBER WINDOWS:

1. HARDWOOD FRAME WINDOWS HAVE LARGER CONTACT AREAS AND SHOULD HAVE A DRAUGHT-REDUCING PROFILE, WITH A CAVITY THAT ALSO SERVES AS A SECONDARY DRIP. THE DRAUGHT PROOFING OF SUCH OPENING SECTIONS IS BY DESIGN MUCH SUPERIOR TO STEEL SECTIONS, WITH BOTH HINGING AND SLIDING-PROJECTING DESIGNS.
2. CLEAN THE CONTACT SURFACE AND APPLY COMPRESSIBLE SEALER STRIPS TO THE VERTICAL SURFACE OF THE FRAME IF SO SPECIFIED OR IN RETROFIT APPLICATIONS.
3. HORIZONTALLY SLIDING HARDWOOD WINDOWS NORMALLY HAVE BRUSH TYPE DRAUGHT PROOFING STRIPS THAT ALSO PREVENT RATTLING.
4. CAREFULLY PROTECT THE SEALER STRIP DURING CONSTRUCTION AND INSPECT FUNCTIONALITY UPON COMPLETION.

Table 2 - Average water consumption (hot and cold) appliances | **Table 5 - Hot water demand, storage and heater power requirements**

1	2	3	4			
Domestic and commercial appliances	L/operation	No of Fittings	Total/L/operation	Total hot water demand	Storage volume at 60°C	Heater power[a]
Bath	80 - 90	2	170			
Bidet	6 - 8					
Clothes washing machine	60 -180	1	100			
Dishwashing machine	3 - 70	1	35			
Domestic waste disposal unit	10 - 15 a					
Shower	3 - 6 a	3	18			
Wash-hand basin	4 - 8	5	25			
WC flushing valve (normal flush)	8 - 10	4	36	460	100	20
Domestic appliances	L/day/person served					
Car washing and garden use	3 - 6	4	20			
Drinking, food preparation and cooking	18 - 22	4	80			
Laundry	10 - 15	4	60			
Personal washing and bathing	20 - 30	4	100			
Washing dishes	8 - 12	4	40			
WC flushing	32 - 40	4	140			
Office installation appliances	L/day/person served					
Hand washing : normal taps	8 - 15					
Hand washing : spray taps	3 - 7					
Urinal flushing - 24 h day	10 - 18					
Urinal flushing - 8 h day	4 - 6					
WC flushing - no urinals provided	12 - 18					
WC flushing - urinals provided	4 - 6					
a Per minute						
		TOTAL	824			

SANS 20252-1 ABBREVIATIONS

MATERIALS

ABS	Alkyl butal styrene
CI	Cast iron
CLPE	Cross-linked polyethylene
CONC	Concrete
COP	Copper
FC	Fibre cement
GMS	Galvanized mild steel
HDPE	High density polyethylene
LDPE	Low density polyethylene
PF	Pitch impregnated fibre
RC	Reinforced concrete
SS	Stainless steel
PVC-u	

FIXTURES AND FITTINGS

BT	Tap (bath)
BTT	Bidet (taps and spouts)
FE	Fire extinguisher
ET	External tap
FH	Fire hydrant
FM	Fire main
FPC	Fire pump connection
FS	Fire switch

FV	Fire valve
HR	Hose reel
HSTO	Hot water storage tank
LCV	Lever control value
NRV	Non-return value
PCV	Pressure control valve
PRV	Pressure reducing valve
PS	pressure switch
PU	Pump
RBT	Registered break tank
RV	Reflux valve
ST	Tap (sink)
STO	Cold water storage tank
SWT	(taps and showerhead)
UA	Urinals-automatically controlled flushing device
UM	Urinals-manually operated flushing device
US	Urinals-sensor-operated flushing device
WCS	Water closet (cistern or automatic shut-off flush valve)
WBT	Tap (wash-hand basin)
WH	Water heater
WTT	Tap (wash trough)
V	Valve

1	2	2	2
Premises	Total hot water demand	Storage volume at 60°C	Heater power[a]
Clinics	(120-150) L/bed/d	(30-35) L/bed/d	1,5 kW/bed/d
Colleges and schools			
Day school	(10-12) L/capita/d	(5-6) L/capita	0,1 kW/capita
Boarding school	(50-115) L/capita/d	(25-50) L/capita	0,5-0,8 kW/capita
Dwelling houses			
Low rental	(80-115) L/capita/d	(100-150) L/capita	2-3 kW/unit
Medium to high rental	(115-140) L/capita/d	(40-50) L/capita	2-5 kW/unit
FactorEX.IEs			
Staff	(10-20) L/capita/d	(5-7) L/capita/d	0,1 kW/capita
Ablutions	(30-60) L/capita/d	(30-60) L/capita/d	1,5-2 kW/capita
Flats (blocks)			
Low rental	(65-75) L/capita/d	(20-25) L/capita	2-3 kW/capita
Medium to high rental	(115-140) L/capita/d	(25-35) L/capita	2-5 kW/capita
Hospitals			
General	(130-140) L/bed/d	(25-30) L/bed/d	1-1,5 kW/bed/d
Infectious	(220-230) L/bed/d	(40-50) L/bed/d	1,5-2 kW/bed/d
InfermarEX.IEs	(65-75) L/capita/d	(20-25) L/capita	0,9-1,2 kW/capita
InfirmarEX.IEs with laundry	(85-95) L/capita/d	(25-30) L/capita	1-1,4 kW/capita
Maternity	(220-230) L/bed/d	(40-50) L/bed/d	1,5-2 kW/bed/d
Mental	(85-95) L/capita/d	(20-25) L/capita/d	1-1,4 kW/capita
Nurses' home	(120-130) L/capita/d	(40-50) L/capita/d	1-1,5 kW/capita
Hostels	(80-120) L/capita/d	(30-35) L/capita/d	0,8-1,1 kW/capita/d
Hotels			
Kitchens	(120-140) L/bed/d	(50-70) L/bed/d	0,9-1,2 kW/bed/d
Full meal preparation	(5-7) L/meal	(5-7) L/meal	0,1 kW/meal
Offices			
Offices with canteens	(25-28) L/capita/d	(20-25) L/capita/d	0,5 kW/capita
Offices without canteens	(10-12) L/capita/d	(5-7) L/capita/d	0,1 kW/capita
Shops (staff only)	(10-12) L/capita/d	(5-6) L/capita	0,1 kW/capita
Sports pavilions Participants only	(30-40) L/capita/d	(30-40) L/capita/d	1,5-2 kW/capita

a Refers to direct electrical heating elements only
b Excluding kitchen but including laundry
c Storage normally a minimum of 15 L with a 4 h heat-up period

GEYSERS

The drip tray of geysers shall be installed and mounted in the following manner:
a) It shall be placed on the tie beams, over a load-bearing wall and be supported by timber that complies with at least grade 5 of SANS 1783-2, and of size at least 114 mm × 50 mm, spaced not more than 500 mm apart and that complies with SANS 10400-L (see figure B.1).
b) It shall be sloped to the outlet end, and rigidly maintained in the position given in (a).
c) The outlet shall be connected to a corresponding drain pipe which shall slope away from the tray and discharge to the outside of the building in a visible position. SANS 10254:2012 B.2.1
When the water heater is
a) mounted direct in the drip tray, each foot shall correspond with the timber specified in B.2.1(a) above, or
b) suspended above the drip tray, the size of the drip tray shall be such that it will cover the total area of the heater and the associated water control appurtenance

b. Vacuum breakers need to be fitted which are small brass components mounted at the end of a 300mm length of pipe. The two vacuum breakers (one for cold and one for hot water) stand vertically about 300mm above the geyser. They stop the water siphoning out the geyser when the cold supply is stopped and they extend the lifetime of the geyser.
c. The geyser must be earthed.
Each exposed conductive part shall be bonded to the cold-water inlet, and earthed.
SANS 10142-1:2012 Edition 1.8 (As amended 2003, 2005 and 2012) 6.16.7.2
d. The geyser must have an isolator in a position where it can be seen and within 1m of the geyser for ease of access and in plain view, secured to a roof truss and not placed on the ceiling or on top of an internal wall.
An electrode water heater, steam generator or boiler shall be connected to the a.c. supply system only. The supply shall be controlled by a multipole circuit-breaker that disconnects all phase conductors and there shall be a switch-disconnector fitted close to, and visible from, the water heater, unless
Amdt 1
a) the circuit-breaker is close to, and visible from, the water heater, or
b) a control switch for isolating the supply is positioned close to the water heater and indicating lights are visible from the control switch, one indicating light showing the closed position of the circuit-breaker and one indicating light showing the open position of the circuit-breaker.
SANS 10142-1:2012 Edition 1.8 (As amended 2003, 2005 and 2012) 6.16.7.1
e. A multi-valve (water pressure regulator) needs to be installed. The valve regulates pressure allowing an even flow of water through the hot and cold taps and is especially important for showers to keep the hot water supply even to sustain the temperature but an even water pressure is also a building regulation.
It is a compulsory requirement in terms of the SABS Code of Practice for geyser installations for both the hot and cold water delivery pressure in the system to be equally balanced. SANS 10254. 4.3.3.
f. An overflow pipe must be fitted as per regulation. The overflow pipe must be 20mm diameter and must be copper or galvanized steel and must be extended to discharge on the outside.
The overflow pipe from either the geyser safety valve or the pressure relief valve may not discharge into the safety tray but they shall be taken outside the building and discharged into an open trapped gully or into a waste pipe designed specifically for that purpose. SANS 10254/2004 / 5.2
The plumber who installs/replaces a water heater or does any maintenance or repair work to the water heater installation, or fixing a leak or a pressure control valve, must first look at the installation and inform the owner or user in writing what aspects of the installation do not comply. The legal responsibility for the safety of the heating system is placed by law on the owner of the property, and who by law must have it fixed.

The legislation requires that all new geyser installations and replacement geyser installations performed after the 8th June 2001 must be in accordance with the specification, which requires the installation to have a geyser drip tray with overflow pipe piped to the exterior of the building – where installed in a roof cavity. The geyser must be earthed and an isolator must be installed close to the geyser.
Vacuum breakers must be installed on the hot and cold water supply and the overflow from the safety valve and expansion relief valve must be piped to the exterior of the building.
13. Plumbing regulations for new or replacement geyser installations make it compulsory for every geyser to have a safety tray installed under the geyser. The size of the safety tray is not only determined by the size of the geyser but also that it shall be large enough to catch any leaks that might occur from any of the geyser valves or ancillary fittings. A drip tray must be installed to collect and discharge water through a pipe outlet to the outside of the building.
14. Most insurance companies reject damage claims relating to geysers if the geyser installation is not according to SANS 10254.

ALL DIMENSIONS TO BE MEASURED OUT AND PEGGED ON SITE BEFORE COMMENCING WORK

ALL REINFORCED CONCRETE SLABS, FOUNDATIONS, COLUMN DETAILS, BEAMS, STAIRS AND RETAINING WALLS TO BE BUILT STRICTLY IN ACCORDANCE WITH ENGINEERS DETAIL AND UNDER HIS/HER SUPERVISION

BUILDING MUST BE IN ACCORDANCE WITH THE APPROVED BUILDING PLAN. NO CHANGES MAY BE MADE TO SUCH PLANS WITHOUT THE EXPRESSED CONSENT OF THE AUTHOR OF SUCH DOCUMENT

FLOORS,WALLS,GEYSER PIPES AND ROOFS TO BE INSULATED IN ACCORDANCE WITH SANS 10400

ARCHITECTURAL DRAWINGS TO BE READ IN CONJUNCTION WITH ENGINEERING DRAWINGS

ACTS OF PARLIAMENT

All Contractors shall ensure that, before any work is put in hand, they comply with all the necessary Acts of Parliament of the Republic of South Africa.

SPECIFICATIONS

1. GENERAL

1.1 THIS DRAWING IS NOT TO BE SCALED, USE FIGURED DIMENSIONS ONLY. ALL DIMENSIONS AND HEIGHTS TO BE CHECKED AND VERIFIED BEFORE ANY WORK COMMENCES ON SITE. ANY DISCREPANCIES SHALL BE REPORTED TO THE ARCHITECT IMMEDIATELY. ALL LEVELS, HEIGHTS OF PLINTHS, DEPTH OF EXCAVATIONS AND NUMBER OF STEPS TO BE FINALLY CHECKED BY CONTRACTOR ON SITE.
1.2 THE SITE TO BE TREATED IN ACCORDANCE WITH S.A.B.S. CODE OF PRACTICE NO. 0124-1977 WITH 'SHELLDRITE' TERMITEPRUFE SOIL POISONER.
1.3 TOP OF FOUNDATIONS TO BE A MINIMUM OF 300mm BELOW NATURAL GROUND LEVEL.
1.4 BACKFILL TO FOUNDATIONS.
1.5 TOP OF 75mm CONCRETE SURFACE BED TO BE MINIMUM OF 150mm ABOVE FINISHED GROUND LEVEL
1.6 75mm THICK CONCRETE SURFACE BED TO BE ON 'GUNPLAS' USB GREEN DAMP PROOF MEMBRANE ON SAND BINDING ON WELL COMPACTED FILL.
1.7 'GUNDLE' BRICKGRIP S.A.B.S. EMBOSSED D.P.C 375 MIC. UNDER ALL WALLS, WINDOW CILLS AND AT CHANGES IN FLOOR LEVELS.
1.8 STORMWATER SHALL BE REMOVED FROM DWELLING, YARD AND SITE.
1.9 ELECTRICAL INSTALLATION SHALL BE 'HEINEMANN' EARTH LEAKAGE SYSTEM IN DISTRIBUTION BOARD.
1.10 ALL BUILDING WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES BUILDING BY-LAWS AND REGULATIONS.

2. FOUNDATIONS

2.1 ALL FOUNDATIONS, RETAINING WALLS, STRUCTURAL CONCRETE WORK AND SUB SOIL STORMWATER DRAINAGE TO CIVIL ENG'S SPECIFICATIONS.
2.2 ALL SOIL COMPACTION TO CIVIL ENG'S SPECIFICATIONS.
2.3 ALL FOUNDATIONS TO BE REINFORCED AS REQUIRED BY ENGINEER.

3. FLOORS

3.1 MIN. 75mm CONCRETE SURFACE BED.
3.2 25mm CEMENT SCREED TO ALL FLOORS.
3.3 SKIRTING TO BE PROVIDED.

4. WALLS

4.1 220mm BRICK WALLS WITH GUNDLE BRICKGRIP DPC.
4.2 BRICKFORCE EVERY 4 COURSES AND EVERY COURSE FOR 4 COURSES OVER OPENINGS
4.3 ALL BRICKWORK MIN. 7 mPa CLASS II.
4.4 WINDOWS BUILT IN WITH GUNDLE BRICKGRIP DPC.
4.5 ALL PAINTWORK TO COMPLY WITH MANUFACTURER'S SPECIFICATIONS WITH NECESSARY UNDERCOATS.
4.6 EXTERNAL-TO PLAN INTERNAL-PLASTER AND PAINT.

ALL LEVELS, DIMENSIONS, DEPTHS OF EXCAVATIONS, HEIGHTS OF PLINTHS, NUMBER OF STEPS AND SIZES OF FOUNDATIONS TO BE PLACED AT A HEIGHT OF EXISTING STRUCTURE

5. ROOF A

5.1 CONCRETE ROOF TILES @ 26° ON 38X38mm GRADED S.A. PINE TIMBER BATTENS @ 320mm C/C MAX ON 225x38mm PRE-FABRICATED, AS PER ENGINEER, S.A.PINE TRUSSES @ 760mm C/C MAX ON 114x38mm TIMBER WALLPLATES. TRUSSES TO BE ANCHORED WITH TWO STRANDS OF 2.5mm GALVANIZED STEEL WIRE FIXED 300mm DEEP IN BRICK WALL. PROVIDE SABS APPROVED PLASTIC UNDERLAY AND SISALATION INSULATION.
5.2 MIN.170mm CONCRETE SLAB TO ENGINEER'S DETAIL AND DESIGN.
5.3 MIN. 55mm VEMICULITE INSULATING SCREED TO 1:60 FALL.
5.4 WATERPROOFING BY SPECIALIST TO MANUFACTURERS SPEC.
5.5 150Ø P.V.C RAINWATER DOWNPIPES FROM FULLBORE OUTLETS.
5.6 SKYLIGHTS TO MANUFACTURERS SPEC.
5.7 GUTTERS : MARLEY STREAMLINE SQUARE PROFILE RAINWATER GUTTERS + DOWNPIPES.
5.8 3mm SKIM COAT RHINOLITE ON PLASTERBOARD CEILING FIXED TO 38x 38mm S.A. PINE BRANDERING AT 450mm C/C FIXED TO TIE BEAMS.

6. WINDOWS AND DOORS

6.1 ANY PANE OF GLASS WHICH IS TO BE INSTALLED WITHOUT SUPPORT FRAME SHALL BE IN ACCORDANCE WITH SABS 0137
6.2 THICKNESS OF PANES IN RELATION TO THEIR AREA SHALL BE IN ACCORDANCE WITH SABS 0137
6.3 EXTERNAL SLIDING DOORS : STANDARD AS PER WINDOWS TO SCHEDULE.
6.4 INTERNAL DOORS : STANDARD HARDWOOD FRAMES TO SCHEDULE
6.5 WINDOWS : TO SCHEDULE
6.6 SLIDING DOORS : DOORS TO SCHEDULE.

7. DRAINAGE NOTES

7.1 100Ø uPVC. SEWER PIPE DRAIN WITH A MIN. FALL OF 1:60.
7.2 100Ø OVP. AT HEAD OF DRAIN PIPE AS SHOWN.
7.3 RODDING EYES AT HEAD OF DRAIN, AT ALL CHANGES OF DIRECTION AND AT MAX. OF 25000 INTERVALS.
7.4 INSPECTION EYES AT ALL JUNCTIONS OF DRAIN, AND TO HAVE MARKED COVERS AT GROUND LEVEL.
7.5 DRAIN PIPE UNDER BUILDINGS TO BE PROTECTED AGAINST LOAD.
7.6 ALL WASTE PIPE TO HAVE 65mm RE-SEAL TRAPS, ALL WASTE PIPES TO BE ACCESSIBLE OVER ENTIRE LENGTH FOR CLEANING AND REPAIRS.
7.7 ALL WASTE PIPES UNDER FLOOR SLAB TO BE SLEEVED.
7.8 ALL SOIL FITTINGS WITH VERTICAL DISCHARGE GREATER THAN 1220 TO HAVE ANTISYPHON VENTPIPES.
7.9 RAINWATER DOWNPIPES TO DISCHARGE A MIN. OF 2440 FROM ANY
7.10 ALL DRAINAGE WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES DRAINAGE BY-LAWS AND REGULATIONS.

Open Studio Designs
Architectural & Interiors
Mobile 083-306-4315
Fax-086-543-8765
openstudio.design@gmail.com

STAND DESCRIPTION
NEW PROPOSED DWELLING ON PTN 465 OF THE FARM KROMDRAAI 560KQ
CLIENT:

WORKING DRAWINGS

TECHNOLOGISTS:	W JUNG-KRUGER
ARCHITECT:	W.R ELLIS
DATE: NOV 18	**DWG NO:**
DRAWN: WJK	**SHEET:** 08 OF 10